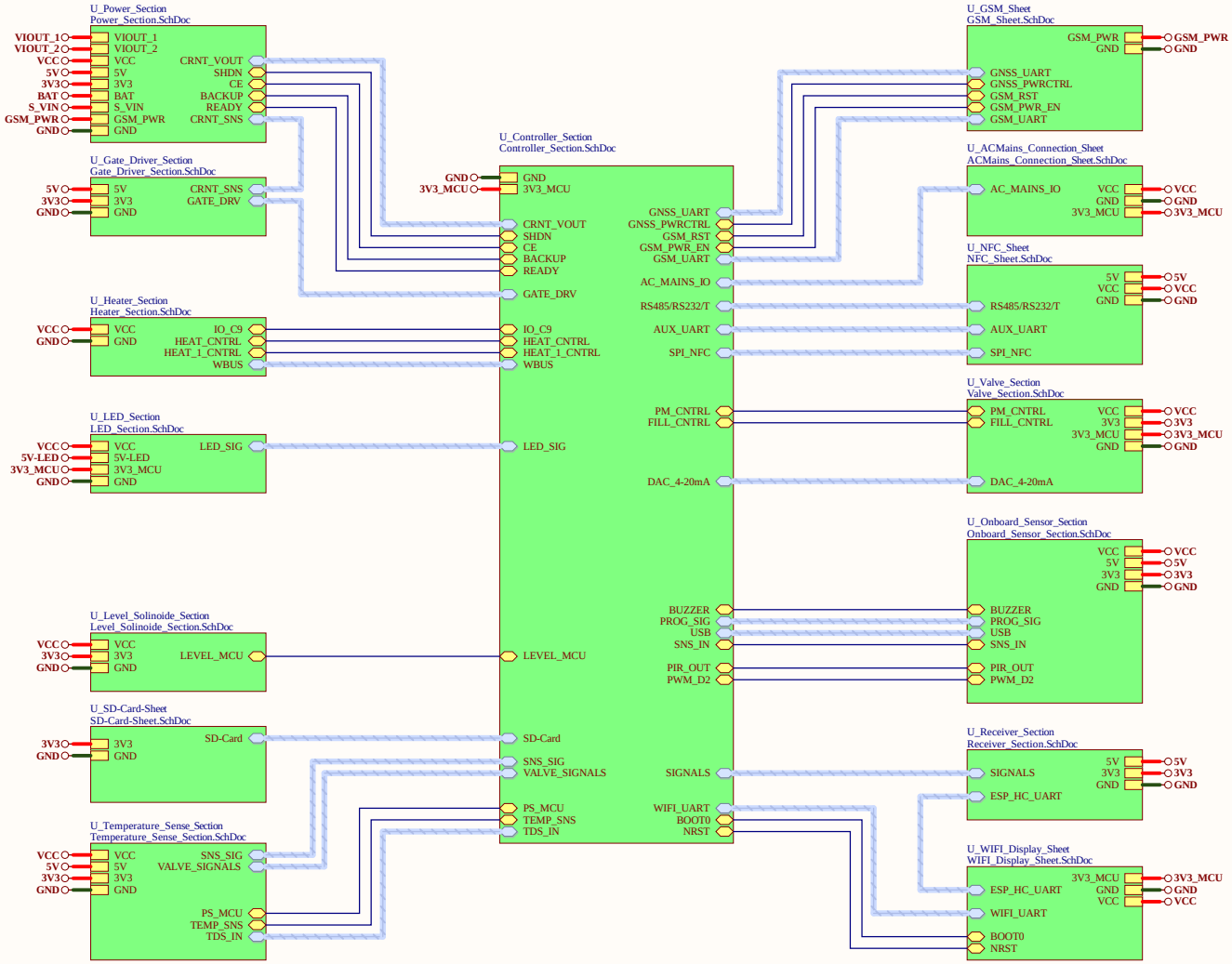
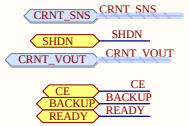
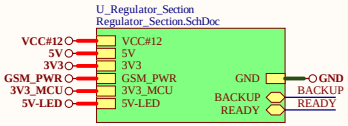
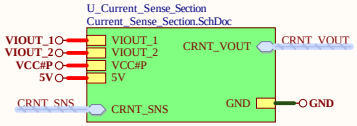


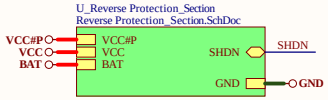
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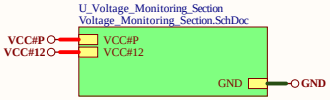
Title: Top_Sheet.SchDoc		
File: GTV4.0_Board.PrjPcb		
Date: 7/22/2025	Page	Num: 1
Drawn By: Huzaifa Ali	Page	Total: 17

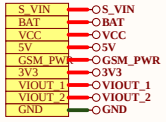


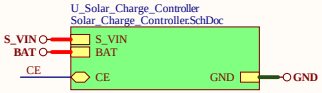








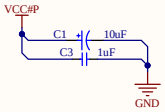




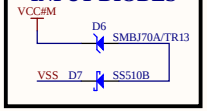
Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium
Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: Power_Section.SchDoc	
File: GTV4.0_Board.PrjPcb	
Date: 7/15/2025	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 17

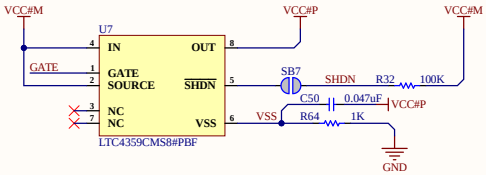
DECOUPLING CAPACITOR



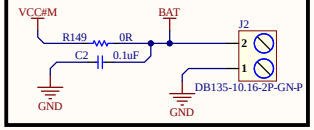
INPUT DIODES



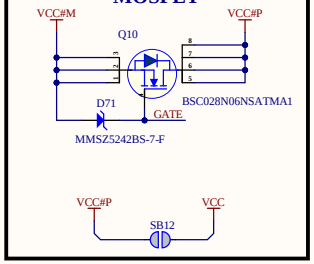
REVERSE POLARITY PROTECTION IC

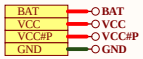


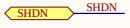
BATTERY / MAIN INPUT



MOSFET



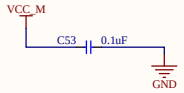




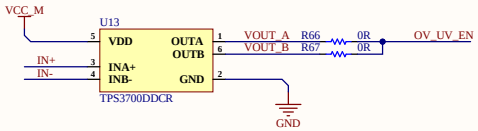
Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Reverse Protection_Section.SchDoc		
File: GTV4.0_Board.PrjPcb		
Date: 11/18/2025	Page	Num: 3
Drawn By: Huzaifa Ali	Page	Total: 17

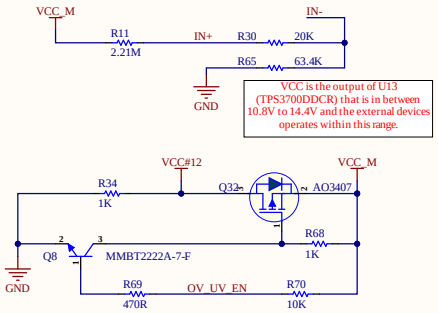
DECOUPLING CAPACITOR



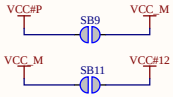
UNDER AND OVER VOLTAGE PROTECTION IC



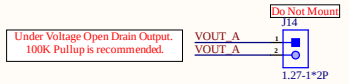
SUPPORTING CIRCUITRY



VOLTAGE TO MONITOR

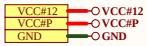


UNDER VOLTAGE OUTPUT



OVER VOLTAGE OUTPUT

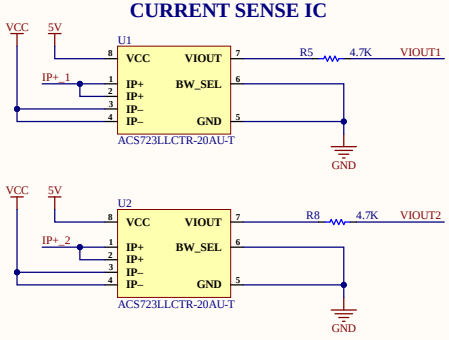




Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

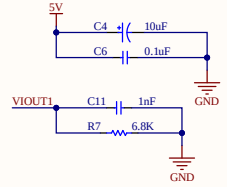
Title: Voltage_Monitoring_Section.SchDoc	
File: GTV4.0_Board.PrjPcb	
Date: 7/15/2025	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 17

CURRENT SENSE IC

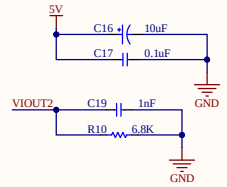


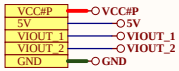
Two current sensing IC's are used to provide support to monitor current of two Pumps. Whereas currently we have only one pump in our V3 Controller.

VOUT1 SUPPORTING CIRCUIT



VOUT2 SUPPORTING CIRCUIT





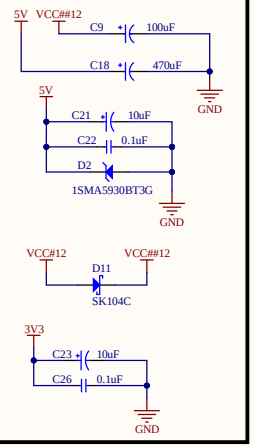




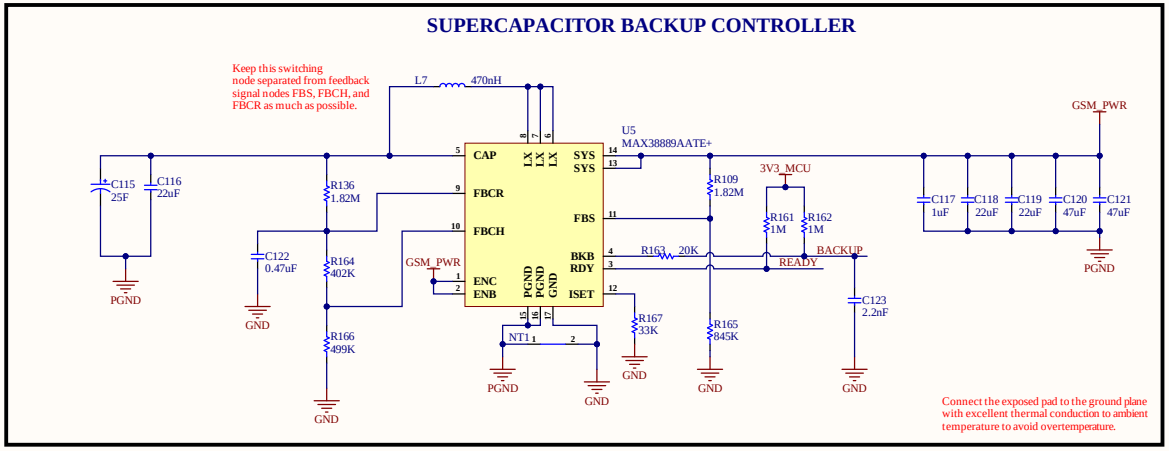
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Title: Current_Sense_Section.SchDoc	
File: GTV4.0_Board.PrjPcb	
Date: 7/15/2025	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 17

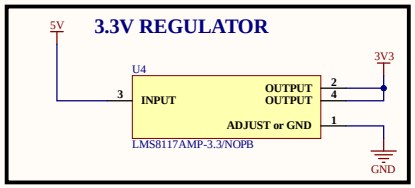
DECOUPLING CAPACITOR



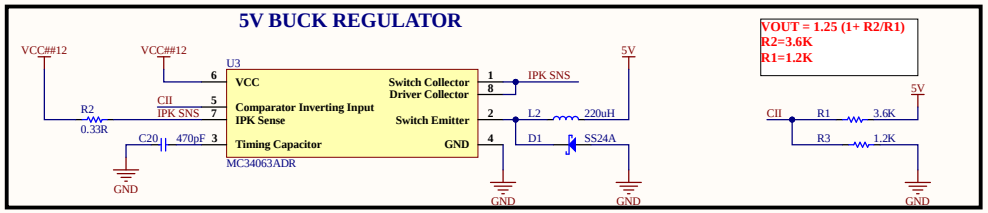
SUPERCAPACITOR BACKUP CONTROLLER



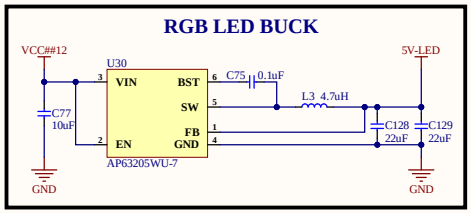
3.3V REGULATOR



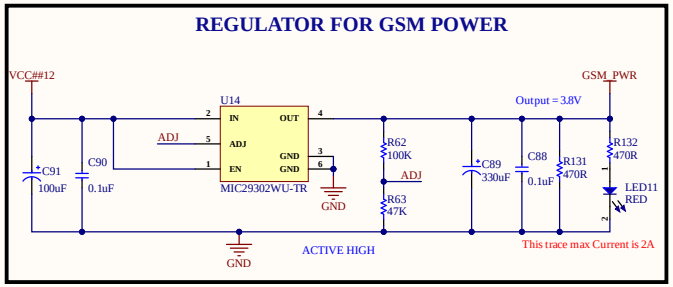
5V BUCK REGULATOR



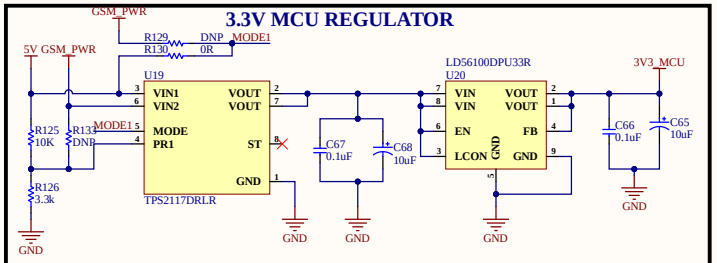
RGB LED BUCK

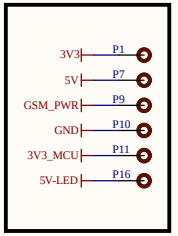


REGULATOR FOR GSM POWER



3.3V MCU REGULATOR



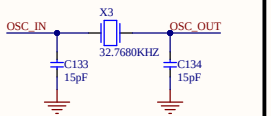


5V-LED
VCC#12
5V
GSM_PWR
3V3
3V3_MCU
GND
BACKUP
READY

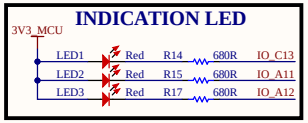
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Title: Regulator_Section.SchDoc	
File: GTV4.0_Board.PrjPcb	
Date: 7/17/2025	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 17

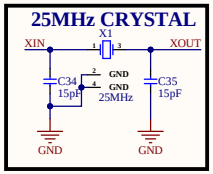
32.768KHZ CRYSTAL



INDICATION LED



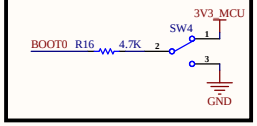
25MHz CRYSTAL



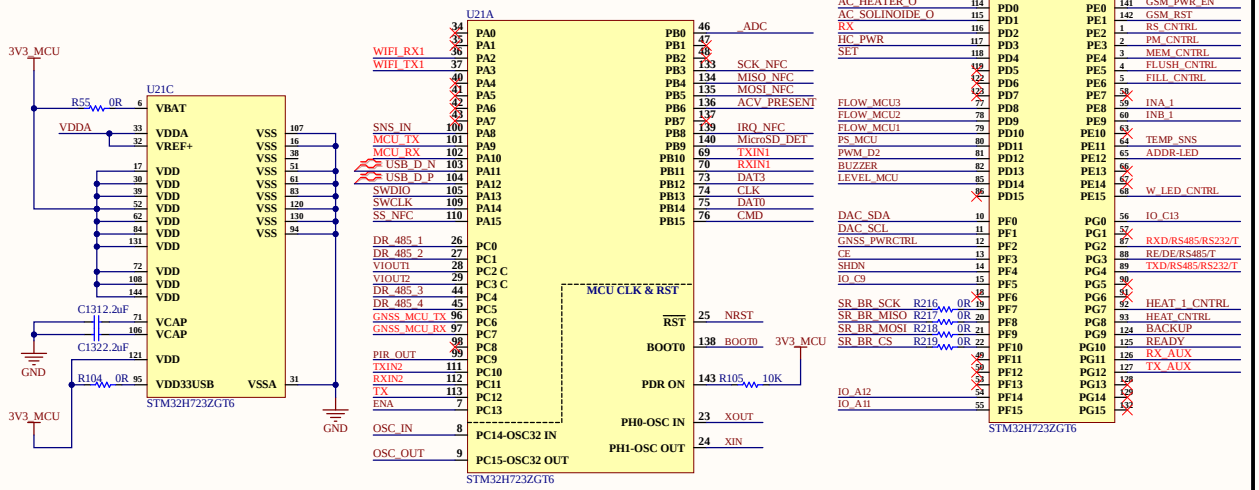
PULL UP



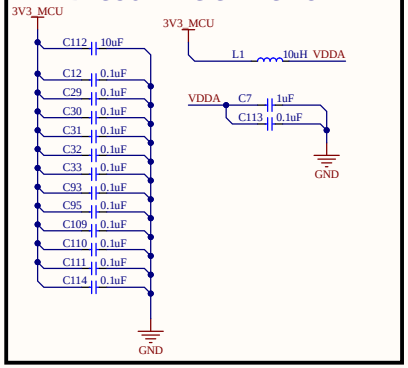
BOOT0 SWITCH



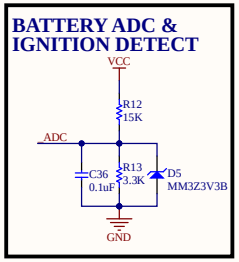
MAIN CONTROLLER



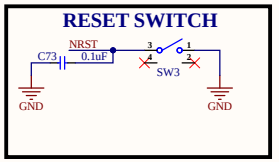
DECOUPLING CAPACITOR

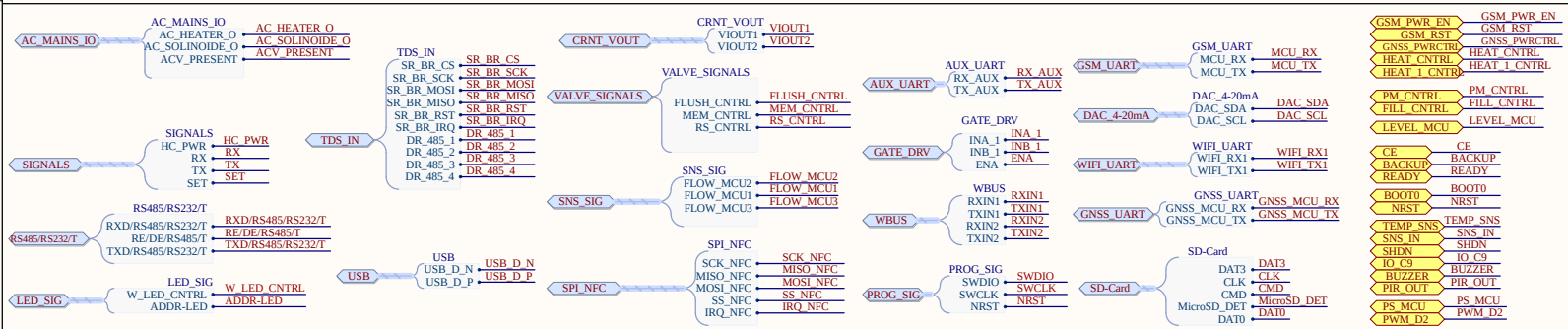


BATTERY ADC & IGNITION DETECT



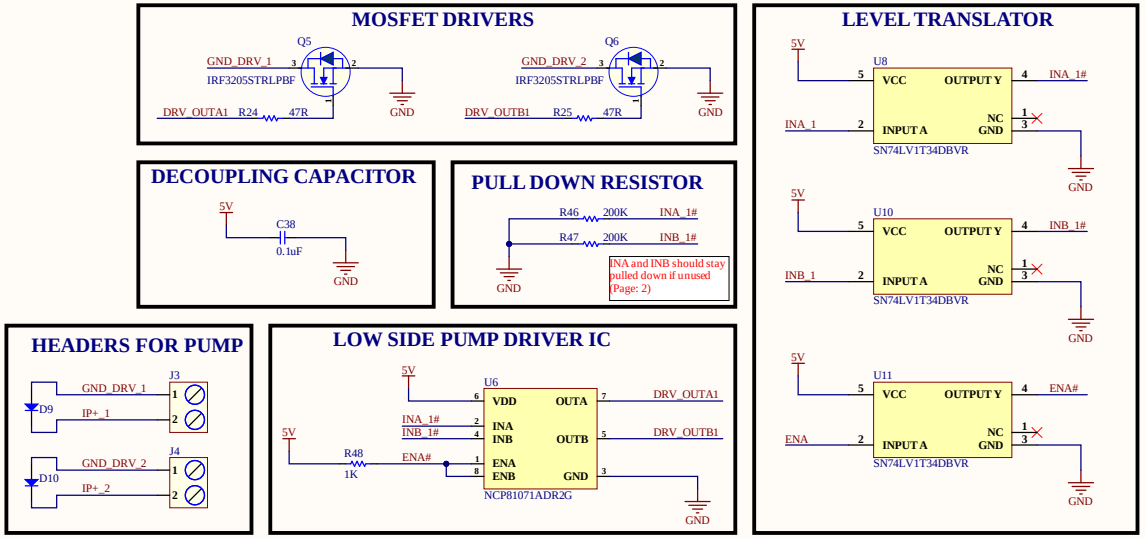
RESET SWITCH





Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Controller_Section.SchDoc	
File: GTV4.0_Board.PrjPcb	
Date: 11/17/2025	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 17



3V3
5V
GND
3V3
5V
GND

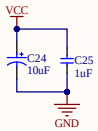
GATE_DRV
GATE_DRV
INA_1
INB_1
ENA
INA_1
INB_1
ENA

CRNT_SNS
CRNT_SNS
IP+_1
IP+_2
IP+_1
IP+_2

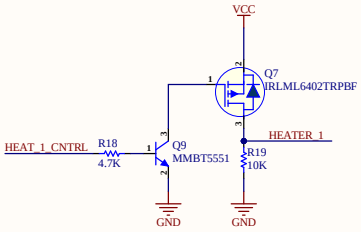
Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Gate_Driver_Section.SchDoc		
File: GTV4.0_Board.PrjPcb		
Date: 7/16/2025	Page	Num: 10
Drawn By: Huzaifa Ali	Page	Total: 17

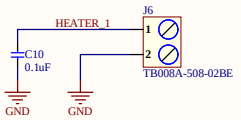
DECOUPLING CAPACITOR

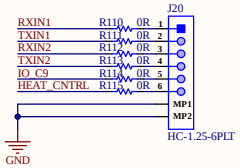


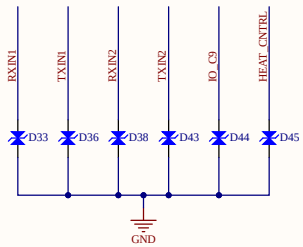
MOSFET CIRCUITRY



HEATER OUT







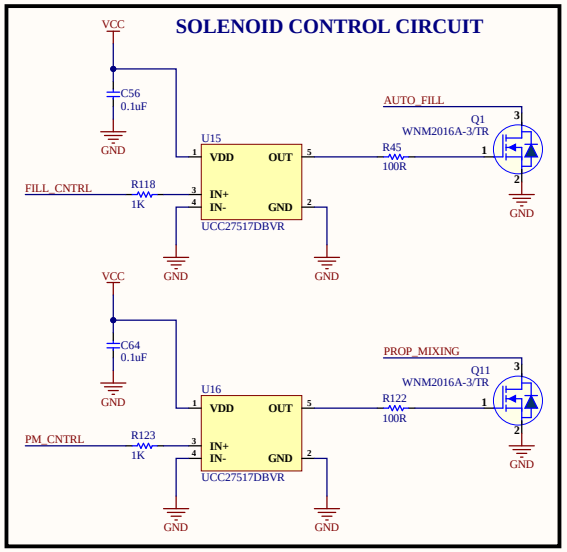


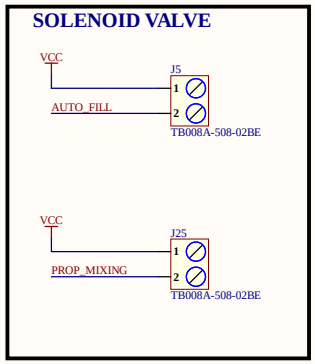


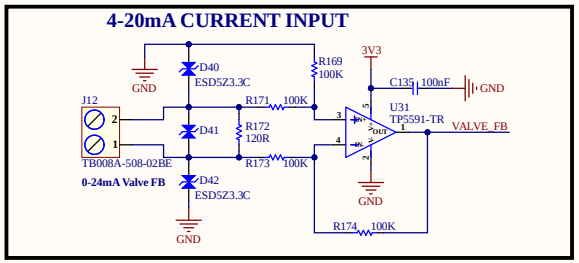


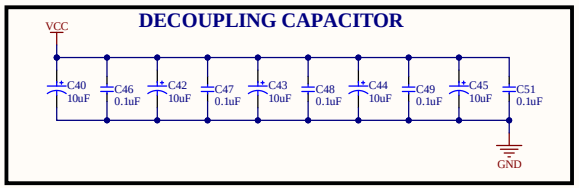
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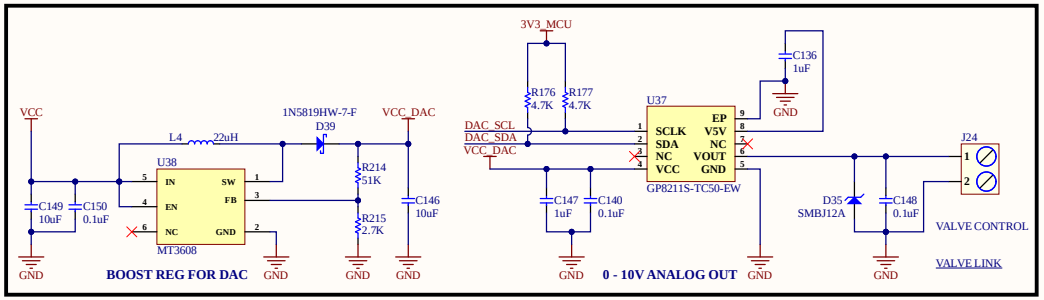
Title: Heater_Section.SchDoc		
File: GTV4.0_Board.PrjPcb		
Date: 11/18/2025	Page	Num: 11
Drawn By: Huzaifa Ali	Page	Total: 17











3V3_MCU — 3V3_MCU
3V3 — 3V3
VCC — VCC
GND — GND

DAC_4-20mA — DAC_4-20mA
DAC_SDA — DAC_SDA
DAC_SCL — DAC_SCL

PM_CNTRL — PM_CNTRL
FILL_CNTRL — FILL_CNTRL

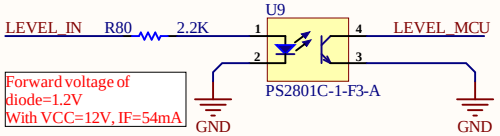
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Title: Valve_Section.SchDoc	
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Date: 11/18/2025	Page Num: 12
Drawn By: Huzaifa Ali	Page Total: 17

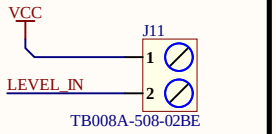
SUPPORTING CIRCUIT

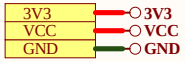


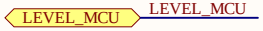
OPTO ISOLATION FOR LEVEL / FLOAT



LEVEL SENSOR INPUT



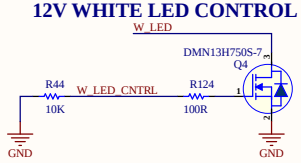




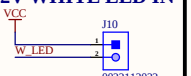
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Date: 7/16/2025	Page Num: 13
Drawn By: Huzaifa Ali	Page Total: 17

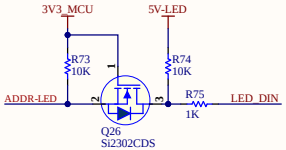
12V WHITE LED CONTROL



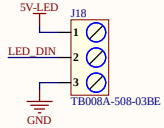
12V WHITE LED IN

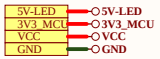


ADDRESSABLE LEDs LEVEL SHIFTER



DIAGNOSTIC LED

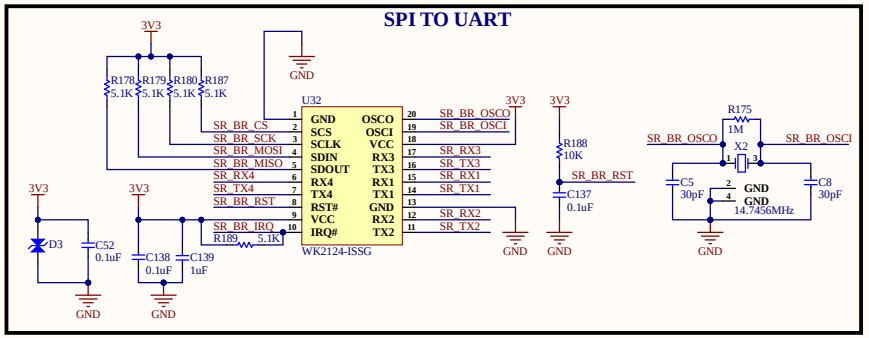


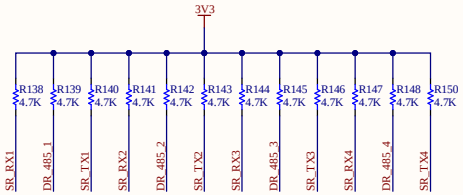


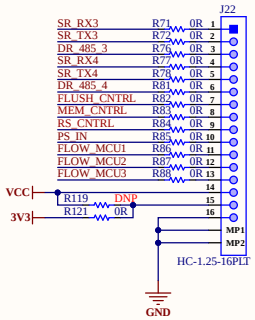


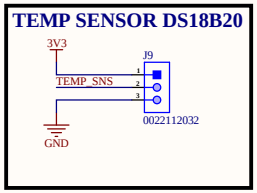
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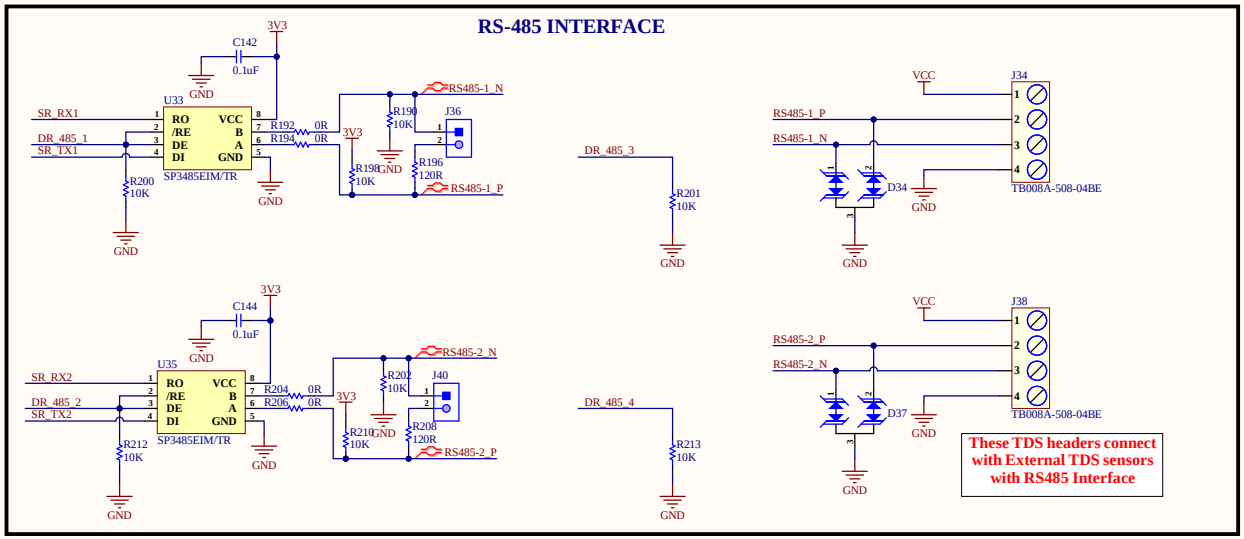
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File: GTV4.0_Board.PrjPcb		
Date: 7/16/2025	Page	Num: 14
Drawn By: Huzaifa Ali	Page	Total: 17

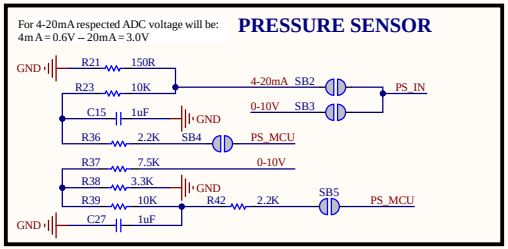


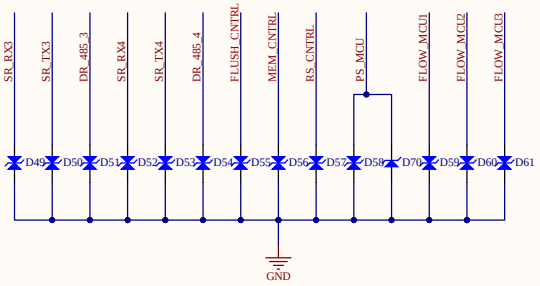


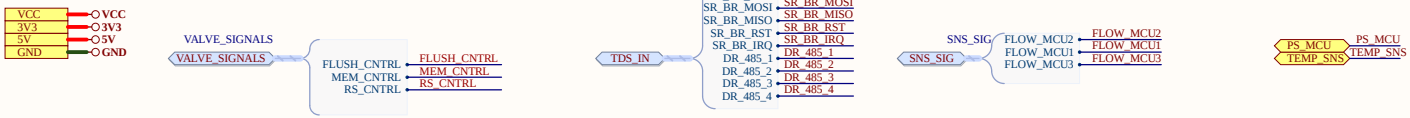








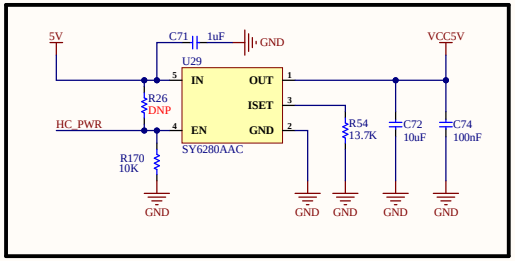




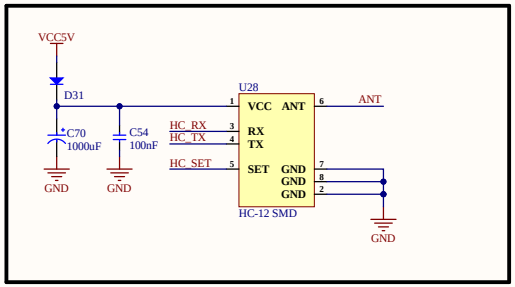
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Title: Temperature_Sense_Section.SchDoc		
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Date: 11/18/2025	Page	Num: 15
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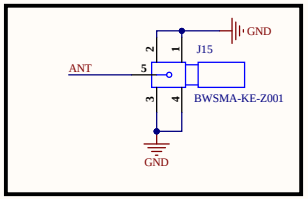
POWER SWITCH



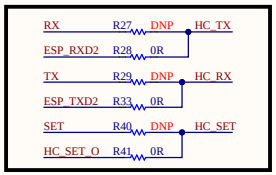
HC-12 433MHz MODULE

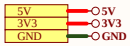


SMA CONNECTOR



CONTROL MCU SELECT





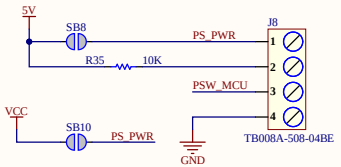




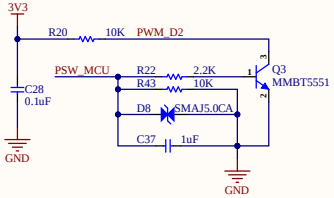
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Date: 11/18/2025	Page Num: 16
Drawn By: Huzaifa Ali	Page Total: 17

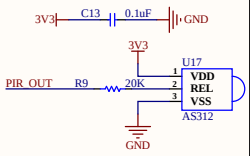
PRESSURE SWITCH



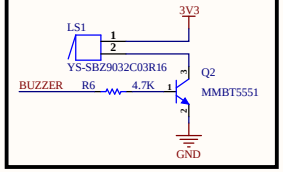
SUPPORTING CIRCUIT



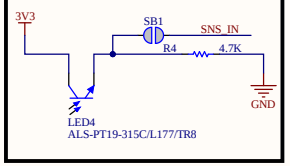
MOTION DETECTION



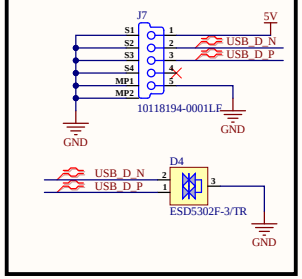
BUZZER CIRCUIT



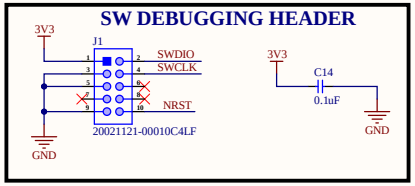
PHOTOTRANSISTOR

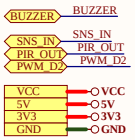


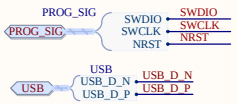
USB TYPE B CONNECTOR



SW DEBUGGING HEADER

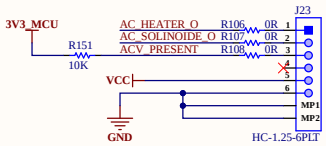


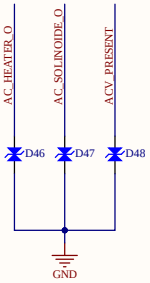




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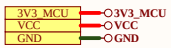
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File: GTV4.0_Board.PrjPcb	
Date: 11/14/2025	Page Num: 17
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MOUNTING HOLES FOR BB PRODUCTION

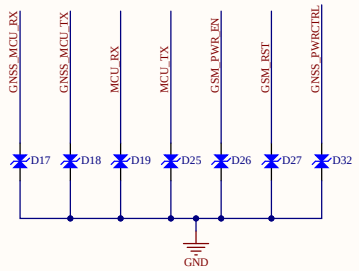


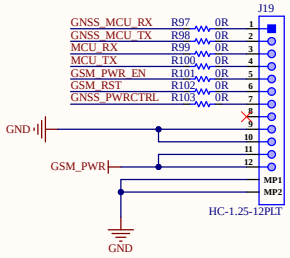




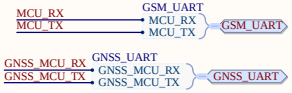
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Title: ACMains_Connection_Sheet.SchDoc	
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Date: 11/17/2025	Page Num: *
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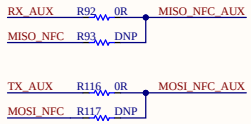


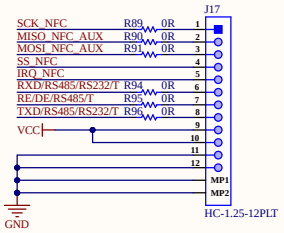


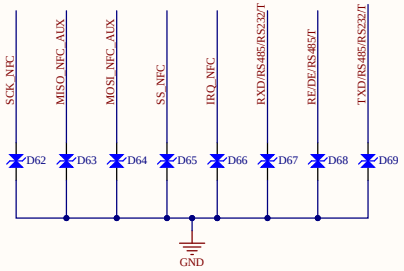


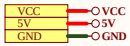
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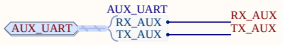
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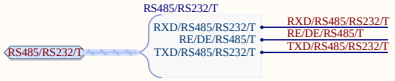






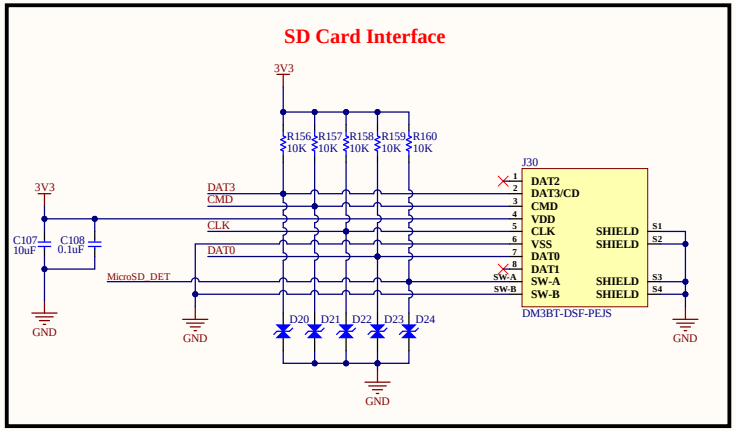






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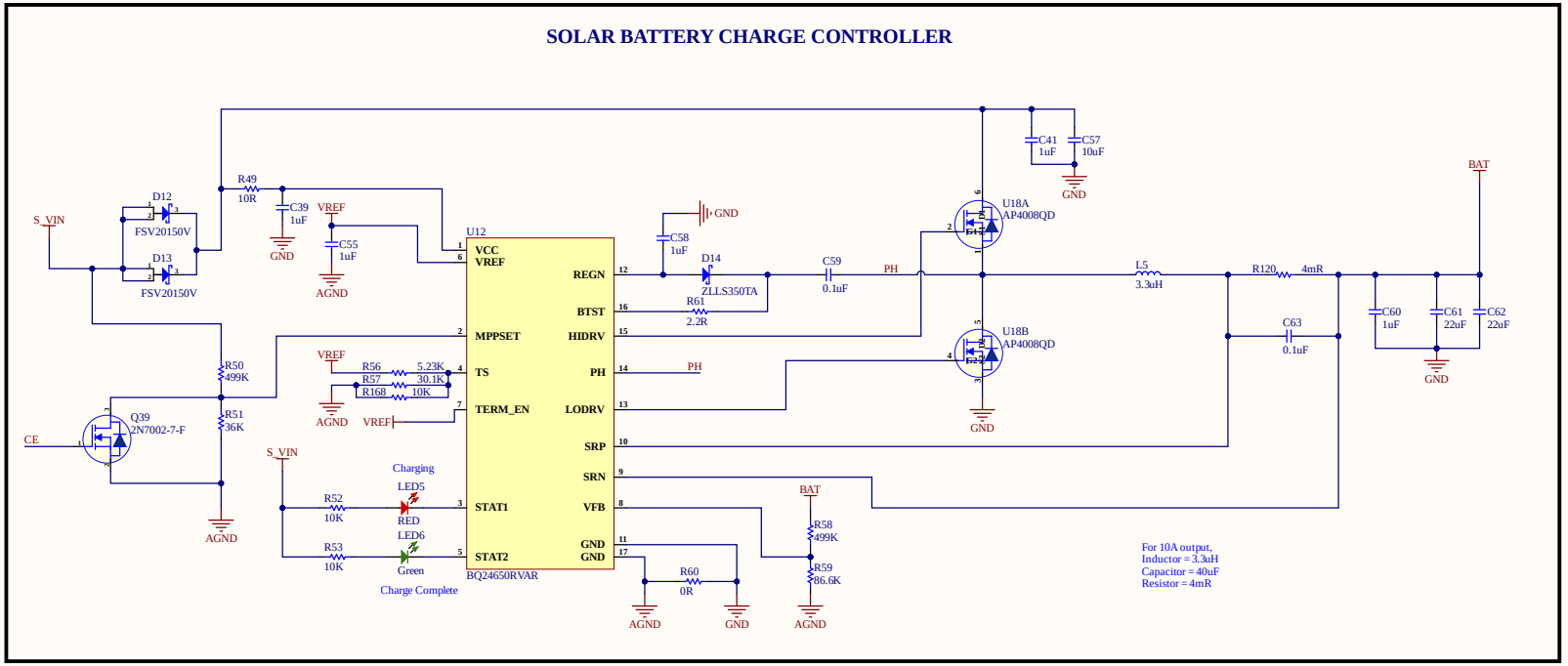
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GND
3V3
GND

SD-Card
SD-Card
DAT3
CLK
CMD
MicroSD_DET
DAT0
DAT3
CLK
CMD
MicroSD_DET
DAT0

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Date: 7/15/2025	Page Num: *
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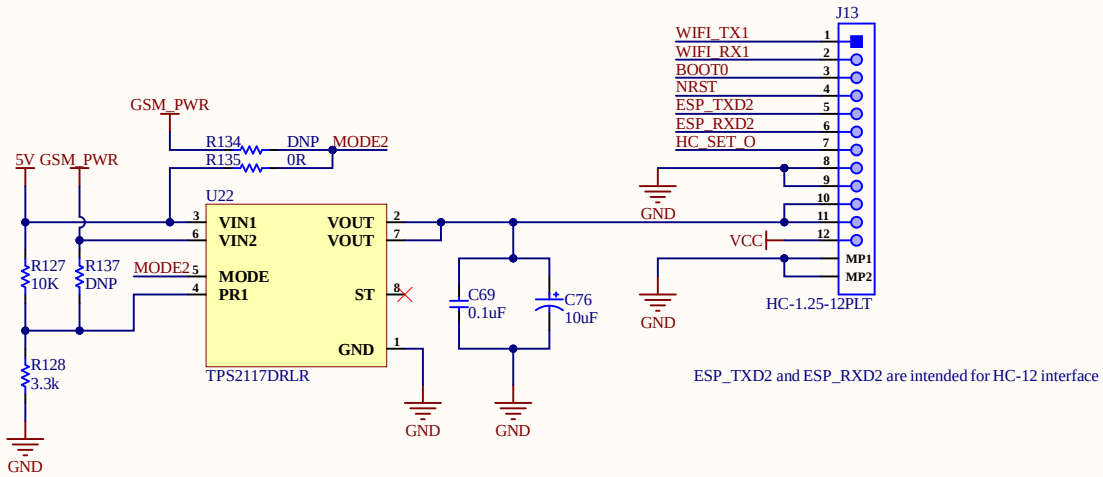
SOLAR BATTERY CHARGE CONTROLLER

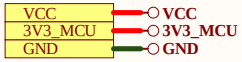


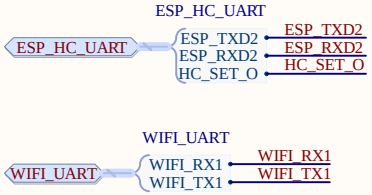
S_VIN
BAT
GND
CE

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Title: Solar_Charge_Controller.SchDoc	
File: GTV4.0_Board.PrjPcb	
Date: 7/16/2025	Page Num: *
Drawn By: Huzaifa Ali	Page Total: *



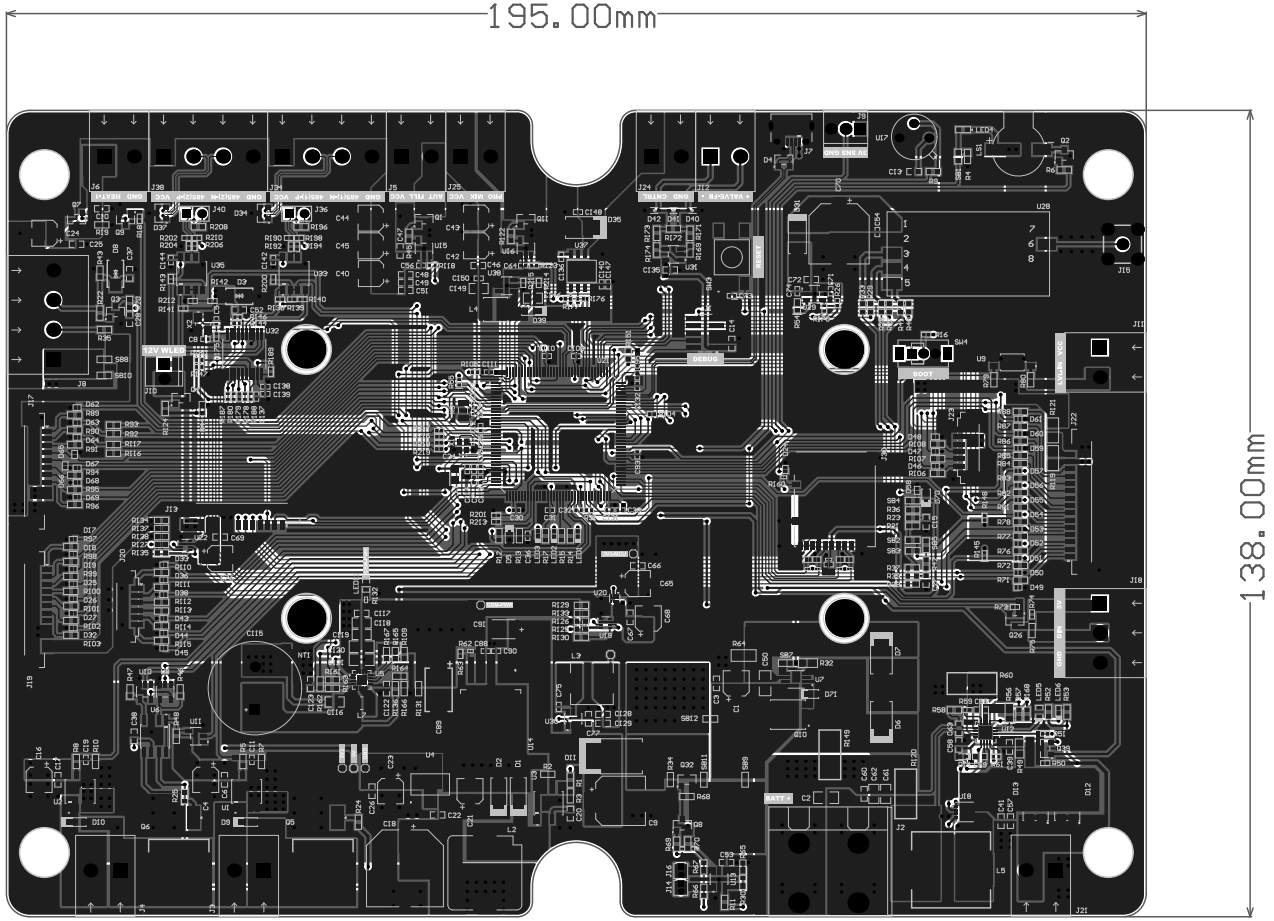






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Title: WIFI_Display_Sheet.SchDoc	
File: GTV4.0_Board.PrjPcb	
Date: 7/17/2025	Page Num: *
Drawn By: Huzaifa Ali	Page Total: *



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.010mm	3.5	
1	Top Layer		0.035mm		
	Dielectric 2	PP-006	0.210mm	4.4	
2	Layer 1	CF-004	0.018mm		
	Dielectric 1	FR-4	1.065mm	4.6	
3	Layer 2	CF-004	0.018mm		
	Dielectric 3	PP-006	0.210mm	4.4	
4	Bottom Layer		0.035mm		
	Bottom Solder	Solder Resist	0.010mm	3.5	
	Bottom Overlay				
Total board thickness:			1.611mm		